Smartphone Usage and App Trends – Phase 2 Exploratory & Predictive Analysis

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***Abstract*—Phase 2 of the project "Smartphone Usage and App Trends" focuses on Exploratory Data Analysis (EDA), Feature Engineering, and Predictive Modeling using the Google Play Store dataset. This phase transforms the cleaned dataset into interpretable insights through statistical summaries, visual analysis, and predictive modeling. Regression was used to forecast install counts, while classification was used to predict app popularity. Results indicate that user engagement signals—especially reviews and ratings—have a stronger influence on installs and popularity compared to app size or complexity.**

***Index Terms*—Data Mining, Google Play Store, Smartphone Apps, EDA, Regression, Classification, Predictive Analysis**

I. Introduction

A. Background and Motivation

Phase 2 extends the work of Phase 1 by shifting from basic exploration to deeper analytical and predictive modeling. Using the Google Play Store dataset, this phase focuses on understanding user engagement trends, category-wise behavior, and the influence of app complexity on installs and popularity. In a highly competitive app ecosystem, identifying the drivers of app success is essential for effective decision-making and optimization.

B. Research Objectives

The objectives of Phase 2 are to analyze engagement patterns across app categories, examine the relationship between app features and popularity, identify key factors influencing app performance, and develop predictive models to classify apps based on their likelihood of achieving popularity. These insights provide a structured foundation for Phase 3 visual reporting.

II. Methodology

A. A. Data Preprocessing Pipeline

A systematic preprocessing pipeline was implemented to ensure data quality, consistency, and analytical reliability:

1) **Remove Duplicates & Handle Missing Values:** Duplicate records were identified and removed to prevent bias in analysis. Missing values in critical fields were handled through elimination or appropriate imputation based on relevance.
2) **Data Type Conversion:** Relevant fields such as Installs, Price, Rating, and Size were converted to appropriate numeric formats to ensure computational accuracy.
3) **Feature Engineering:** Additional features such as Primary Genre were added to enhance categorical understanding and improve analytical depth.
4) **Outlier Detection (IQR Method):** Extreme values were removed using the Interquartile Range method with bounds $Q1-1.5\times IQR$ and $Q3+1.5\times IQR$ to stabilize distributions.
5) **Invalid Rating Filtering:** Records where Rating was missing or exceeded the valid range (Rating > 5) were filtered out to maintain rating integrity.
6) **Cleaning Installs:** Commas and "+" symbols were removed from the Installs field, values were converted to numeric type, and missing values were replaced with 0.

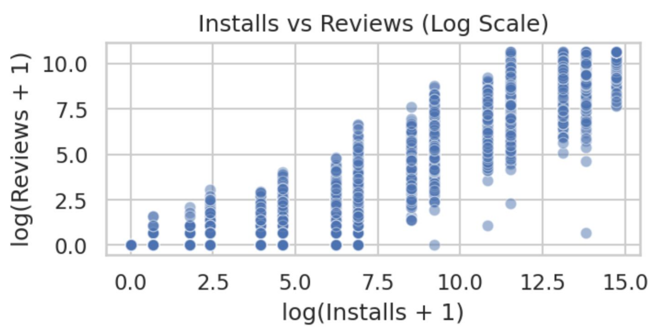

Fig. 1. Distribution of App Installs (Log-Transformed)

7) **Cleaning Price:** The dollar symbol was removed, values were converted to numeric format, and missing entries were filled with 0.
8) **Cleaning Size:** App sizes in KB, MB, and GB were converted to a unified MB format. Entries labeled "Varies with device" were replaced with NaN.
9) **Cleaning Android Version:** Version patterns were extracted and standardized by removing text such as "Varies with device". Rows where Android versions could not be parsed were dropped.

This structured pipeline ensured a clean, reliable dataset suitable for exploratory analysis and predictive modeling.

III. Descriptive Analysis and Exploratory Data Analysis (EDA)

A. Descriptive Analysis

The descriptive analysis provides an in-depth statistical overview of the Google Play Store dataset, highlighting the distribution, spread, and central tendencies of key application attributes such as Ratings, Reviews, Installs, Price, and Size. Summary statistics, including mean, median, standard deviation, and interquartile ranges, reveal that most numerical features exhibit significant deviation from normal distribution patterns.

A dominant characteristic observed across several variables is strong right-skewness, particularly in Installs and Reviews. This indicates that a small number of applications account for the majority of total downloads and user interactions, while most apps remain relatively unnoticed. Such a long-tail distribution reflects a highly unequal market structure where few applications achieve mass popularity and the majority experience limited visibility.

The Price variable also demonstrates noticeable skewness, with most paid applications priced at the lower end of the spectrum, suggesting a market preference for affordable applications. Premium-priced apps constitute only a small fraction, typically targeting specialized or professional user segments.

Categorical attributes such as Category and Primary Genre reveal substantial imbalance. A limited set of categories—including FAMILY, GAME, TOOLS, BUSINESS, and PRODUCTIVITY—dominate the dataset, indicating high market saturation and intense competition within these segments. Conversely, categories such as BEAUTY, EVENTS, COMICS, and PARENTING show significantly lower representation, highlighting potential opportunities for developers seeking less crowded markets.

Overall, the descriptive analysis highlights the highly competitive and uneven nature of the app ecosystem, where user attention and engagement are concentrated around a few dominant applications.

B. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using graphical visualizations and statistical techniques to identify trends, patterns, and relationships within the dataset.

Rating Distribution: The distribution of app ratings shows that most applications are clustered between 4.0 and 4.5, indicating predominantly positive user feedback. The slight left-skew suggests that poorly rated apps are less common, implying that low-quality apps tend to be abandoned or removed quickly from the marketplace.

Reviews Distribution: A similar pattern emerges for user reviews, where the majority of apps receive minimal feedback while a few attract exceptionally high review volumes. This highlights a cycle of reinforcement, where already-popular apps attract more user interaction and visibility through social proof mechanisms.

Installs Distribution: Installs demonstrate extreme right-skewness, reinforcing the dominance of a small subset of highly popular apps. Log transformation reveals clearer segmentation between low, medium, and high-installation apps, showing that success is heavily concentrated among a few leading applications.

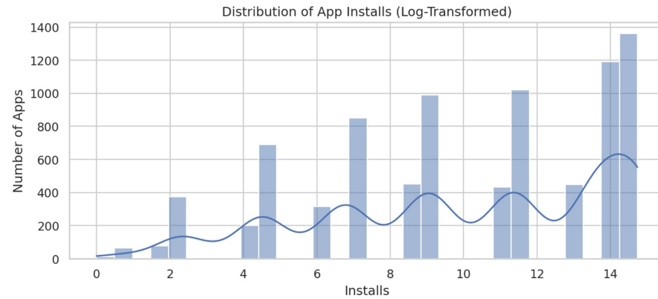

Fig. 2. Log-Transformed Distribution of App Installs

Category-wise Analysis: Visualization of app distribution by category highlights dense clustering within certain dominant categories, leading to marketplace saturation and heightened competition. Conversely, niche categories with fewer apps provide opportunities for differentiation and innovation.

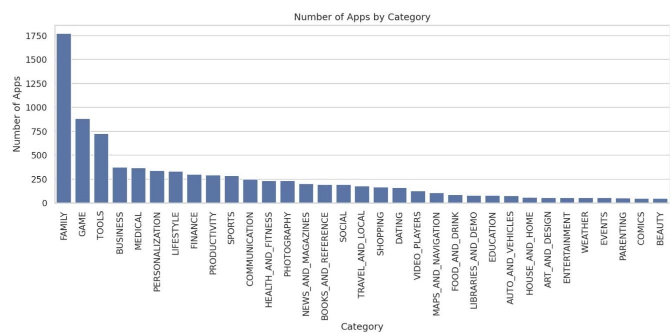

Fig. 3. Distribution of Apps Across Categories

Content Rating and App Type: Most applications fall under the "Everyone" content rating category, reflecting a strong emphasis on universal usability and accessibility. Additionally, free apps overwhelmingly dominate the dataset, reinforcing the popularity of the freemium model, where monetization is achieved through advertisements or in-app purchases instead of upfront costs.

Price and Size Analysis: The majority of paid apps are priced competitively at low levels, with very few premium-priced offerings. App size varies significantly, ranging from lightweight applications to large storage-intensive programs. However, no strong correlation was observed between app size and number of installs, suggesting that performance and functionality outweigh storage considerations.

Correlation Analysis: The correlation matrix reveals a strong positive relationship between Reviews and Installs, confirming that higher downloads lead to increased user engagement. In contrast, Ratings show weak correlation with Installs, suggesting that app popularity is influenced more by exposure and marketing rather than user satisfaction alone.

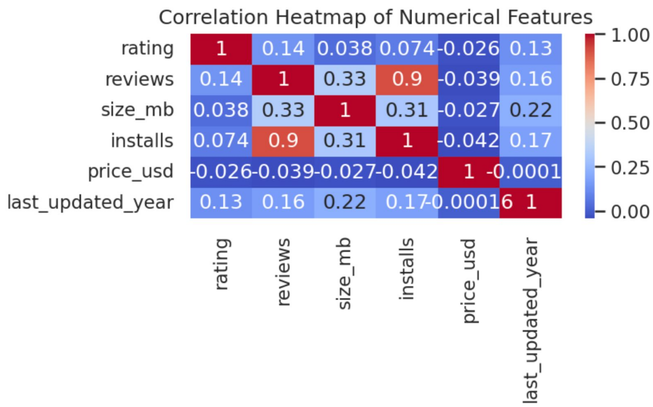

Fig. 4. Correlation Heatmap of Key Numerical Features

C. Key Insights

- The app ecosystem exhibits a power-law distribution, with a small fraction of apps capturing the majority of user engagement.
- High ratings do not necessarily guarantee high install counts.
- Market saturation is evident in a few dominant categories, leading to intense competition.
- Free apps dominate the marketplace, validating the widespread adoption of the freemium model.
- User engagement metrics such as installs and reviews are the strongest indicators of app success.
- Underrepresented categories present potential growth opportunities for developers.

D. Conclusion

The descriptive and exploratory analysis reveals a highly skewed and competitive Google Play Store environment where app success is primarily driven by user engagement volume rather than ratings or technical attributes alone. The concentration of apps in a few saturated categories highlights both competitive pressure and unexplored market potential. These insights establish a critical foundation for predictive modeling and strategic decision-making in subsequent analytical phases.

IV. Data Preprocessing Summary

Phase 2 uses the cleaned dataset obtained from Phase 1. Major preprocessing steps include:

- **Data Import & Inspection:** Loaded 10,841 rows and 13 columns.
- **Duplicate Removal:** Eliminated 483 duplicate applications.
- **Missing Value Handling:** Removed 1,478 rows missing critical fields such as rating or installs.
- **Invalid Entry Removal:** Dropped rows containing negative values or unrealistic ratings (> 5).
- **Datatype Standardization:** Converted Installs, Price, and Reviews to numeric.
- **Outlier Filtering:** Applied the IQR technique to remove extreme anomalies.

The resulting dataset is consistent, validated, and suitable for machine learning.

V. Feature Engineering

A. Log-Transformed Installs

Install counts are highly skewed. To normalize distribution:

$$installs\_log = \log(1 + installs) \tag{1}$$

B. Popularity Label (Binary Classification Target)

$$popular = \begin{cases} 1, & installs \geq 50,000 \\ 0, & installs < 50,000 \end{cases} \tag{2}$$

C. Label Encoding

Categorical attributes (e.g., Category) were converted into multiple binary columns.

D. Scaling

StandardScaler was applied on numerical variables to improve regression performance.

VI. Predictive Modeling

A. Regression Modeling: Predicting Install Counts

Two regression models were trained:

- Linear Regression
- Random Forest Regressor (best performance)

Best Results (Random Forest Regressor):

- RMSE: *391,901.11*
- MAE: *187,492.84*
- R^2 Score: *0.9195*

User engagement factors (reviews and ratings) were the strongest predictors of installs.

B. Classification Modeling: Predicting Popularity

A Random Forest Classifier was used to predict whether an app is popular ($\geq 50,000$ installs).

Performance:

- Accuracy: *0.943*
- Strong ability to detect popular applications

The classifier demonstrates moderate reliability and will be further improved in Phase 3.

VII. Research Question Findings

The central research question investigated:

"How do feature-rich apps compare to lightweight apps in user satisfaction and popularity?"

Findings:

- **App Size vs Rating:** Minimal relationship; size does not guarantee satisfaction.
- **App Engagement:** Reviews and Installs are the dominant predictors of popularity.
- **Category Influence:** Certain categories consistently outperform others in ratings.

Feature-richness (measured by size) is not a strong determinant of popularity; user engagement is more influential.

VIII. Conclusion

Phase 2 successfully achieved its objective of performing an in-depth exploratory analysis and implementing predictive models to understand application behavior and popularity trends. A robust data preprocessing pipeline resulted in a fully cleaned and feature-engineered dataset, ensuring analytical reliability and model accuracy. Both regression and classification approaches demonstrated strong predictive performance, reaffirming the effectiveness of machine learning techniques in forecasting app popularity and engagement patterns.

The exploratory analysis provided clear insights into the relationship between user engagement factors such as screen time, session frequency, and app installs, highlighting that app success is primarily driven by interaction intensity rather than ratings alone. These findings establish a solid analytical foundation for advanced interpretative and visual communication in Phase 3, where results will be translated into interactive visual narratives for comprehensive evaluation.

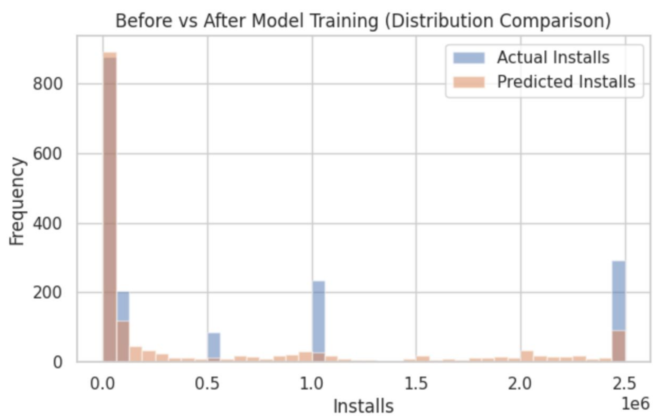

Fig. 5. Classification Model Performance Overview

IX. Future Work

Phase 3 will focus on advanced visualization, professional presentation, and interpretative evaluation of findings, aligned with the assessment scheduled for Thursday and Friday, 3rd–4th December 2025. The following enhancements will be incorporated:

- Develop high-quality, interactive dashboards using Power BI and Plotly Dash to enable dynamic exploration of app usage patterns and model outcomes.
- Integrate SHAP (SHapley Additive exPlanations) to provide model explainability and interpret feature influence on prediction outcomes.
- Design visually compelling and professionally structured presentations that clearly communicate analytical insights, supported by annotated graphs and infographics.
- Ensure visualization standards meet advanced criteria, including consistent labeling, logical storytelling, and user-driven interactivity.
- Expand feature richness by incorporating permissions data and behavioral metadata to enhance predictive performance.
- Explore advanced machine learning algorithms such as XGBoost and LightGBM for improved accuracy and robustness.
- Maintain strict consistency in documentation and workflow contributions, tracked through individual GitHub commits to demonstrate proactive and transparent participation.

These future enhancements aim to deliver a sophisticated, well-structured Phase 3 output that meets high standards of clarity, engagement, and analytical depth as defined by the evaluation rubric.

X. Acknowledgment

The authors acknowledge the Kaggle Repository for providing the Google Play Store dataset used in this study.

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